

Name: _____

Date: _____

Course: CS 471 - Artificial Intelligence

Lesson 15: Reinforcement Learning III (Q-Learning)

Section 1: Instructor-Led Walkthrough

Objective: Calculate a Q-Value update for an autonomous perimeter sentry.

Given the Q-learning update equation:

$$Q_{k+1}(s, a) \leftarrow (1 - \alpha)Q_k(s, a) + \alpha \left[R(s, a, s') + \gamma \max_{a'} Q_k(s', a') \right]$$

Assume the sentry is at Checkpoint 1 (s), takes action "Advance" (a), and receives a reward of $R = -2$. It transitions to Checkpoint 2 (s'), which has a maximum future Q-value of 10. Let $\alpha = 0.5$, $\gamma = 1.0$, and the initial $Q(s, a) = 0$.

Step 1: Calculate the sample

$sample =$

Step 2: Calculate the new Q-value

$Q(s, a) \leftarrow$

Section 2: Student Practice

Problem 1: Exploration vs. Exploitation

An Electronic Warfare algorithm utilizes an ϵ -greedy strategy to find optimal frequencies for jamming adversary comms. If $\epsilon = 0.1$, explain the behavioral tradeoff the algorithm makes at each time step.

Problem 2: Model-Free Learning

Why does an autonomous drone using Q-Learning not require prior knowledge of the transition function $T(s, a, s')$ to learn an optimal flight path?